

EVENTGUARD

UML & Architecture Diagrams

Document Type: Technical Reference

Contains: Use Case, Class, Sequence, Activity (x2), Deployment Diagrams

Version: 1.0 | Ref: SRS Appendix E

UML and Architecture Diagrams

This document contains all six UML and architecture diagrams for the EventGuard platform. Each diagram is presented on a dedicated page with a title bar, a contextual description, and an italic figure caption. These diagrams are extracted from SRS Appendix E.

Figure 1 — Use Case Diagram

Figure 2 — Class Diagram (Domain Model)

Figure 3 — System Sequence Diagram

Figure 4 — Activity Diagram — Risk Assessment & Subscription

Figure 5 — Activity Diagram — Claims Filing & Adjudication

Figure 6 — Deployment Diagram

1. Use Case Diagram

Captures the full functional scope across Customer, Underwriter, Claims Officer, and System Admin. Use cases are grouped within the EventGuard System boundary with <<include>> relationships for shared sub-functions: Risk Calculation, Weather Data Fetch, and Upload Evidence.

Figure 1 — Use Case Diagram

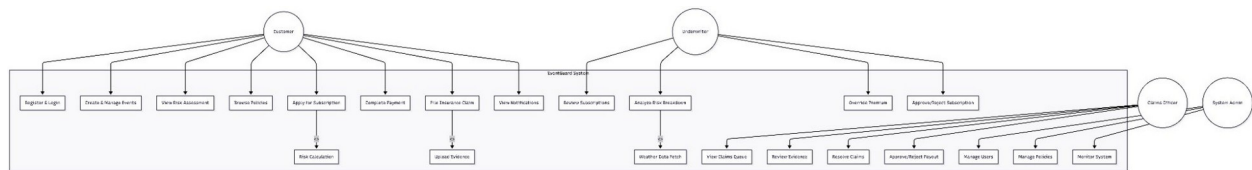


Figure 1 — Use Case Diagram

2. Class Diagram (Domain Model)

Shows all primary Java entity classes (User, Event, Policy, PolicySubscription, Risk, Claim, Notification) with attributes, data types, and inter-class associations. Enumeration types (EventDomain, VenueType, SubscriptionStatus, ClaimStatus) are shown with multiplicities.

Figure 2 — Class Diagram (Domain Model)

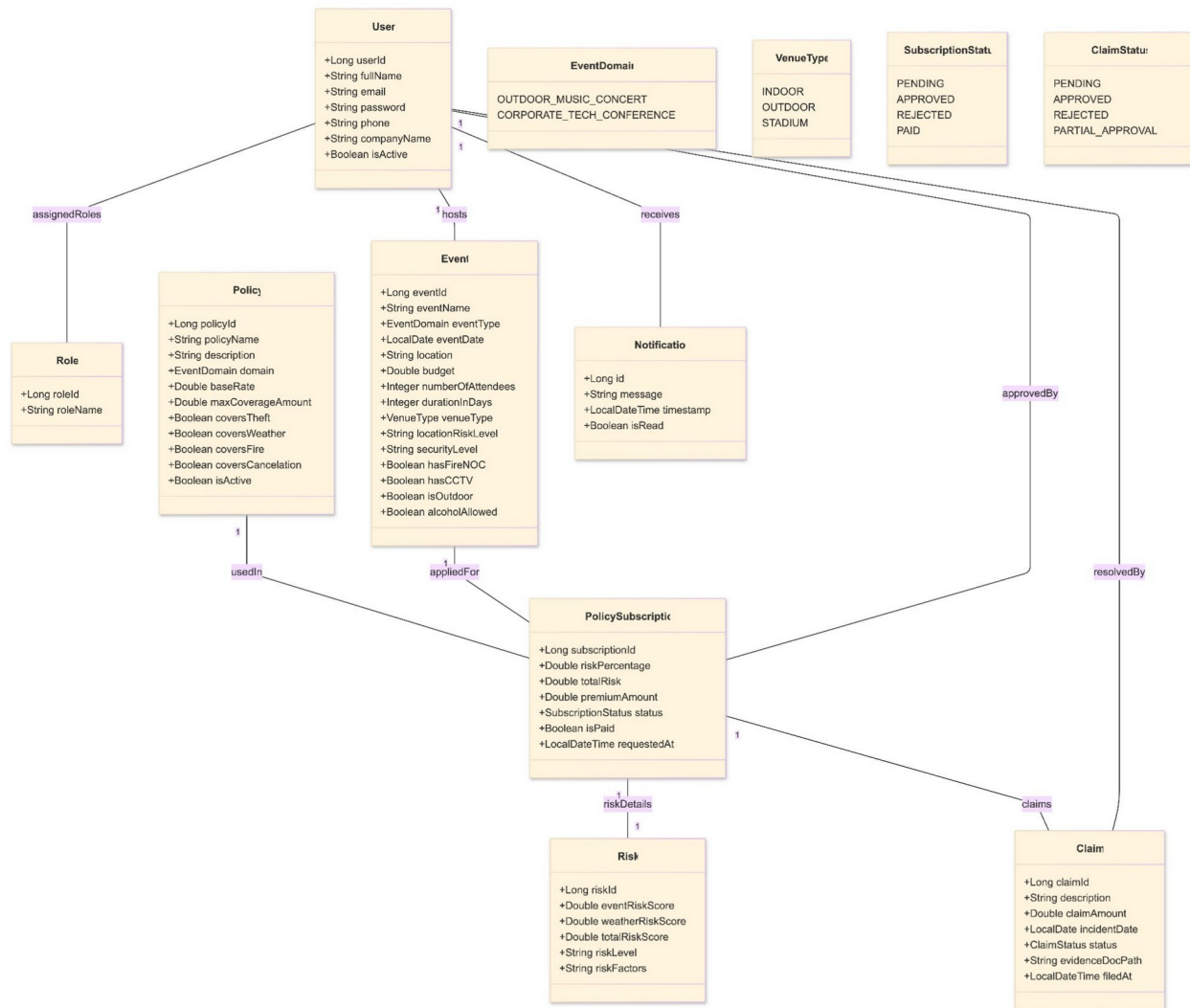


Figure 2 — Class Diagram (Domain Model)

3. System Sequence Diagram

Full end-to-end message flow across: Customer, Web Portal (Angular 21), API Gateway/Auth, Event & Risk Service, WeatherAPI, Insurance Underwriter, Claims Officer, and H2 Database. Covers login, event creation, risk calculation, subscription, payment, claims, and notifications.

Figure 3 — System Sequence Diagram

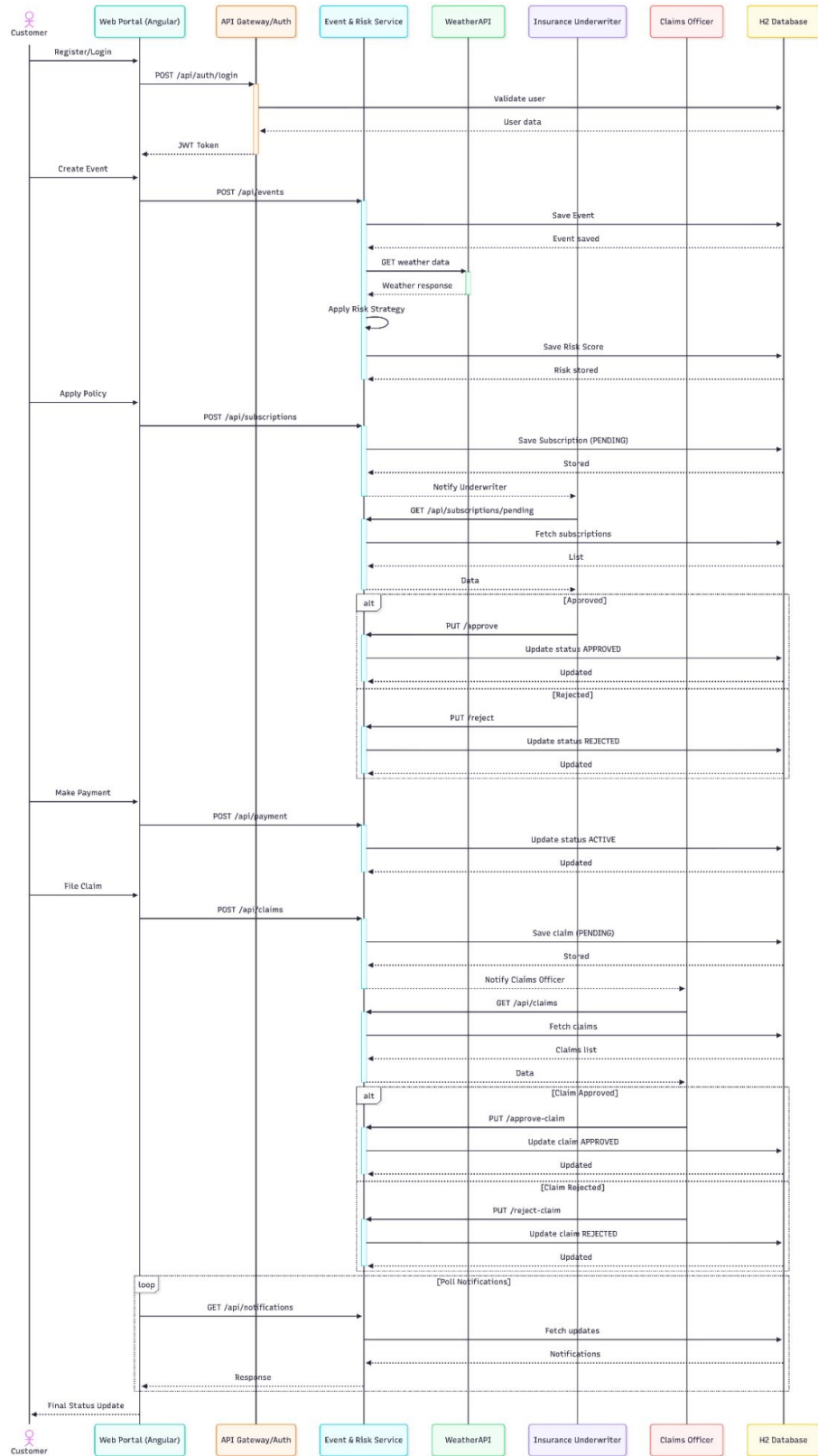


Figure 3 — System Sequence Diagram

4. Activity Diagram — Risk Assessment & Subscription

Models the flow from event detail entry through conditional WeatherRiskService invocation (decision node s11 for outdoor events), domain penalty/mitigation application via the Strategy Pattern, underwriter approve/reject decision (s12), payment, and Subscription ACTIVE status.

Figure 4 — Activity Diagram — Risk Assessment & Subscription

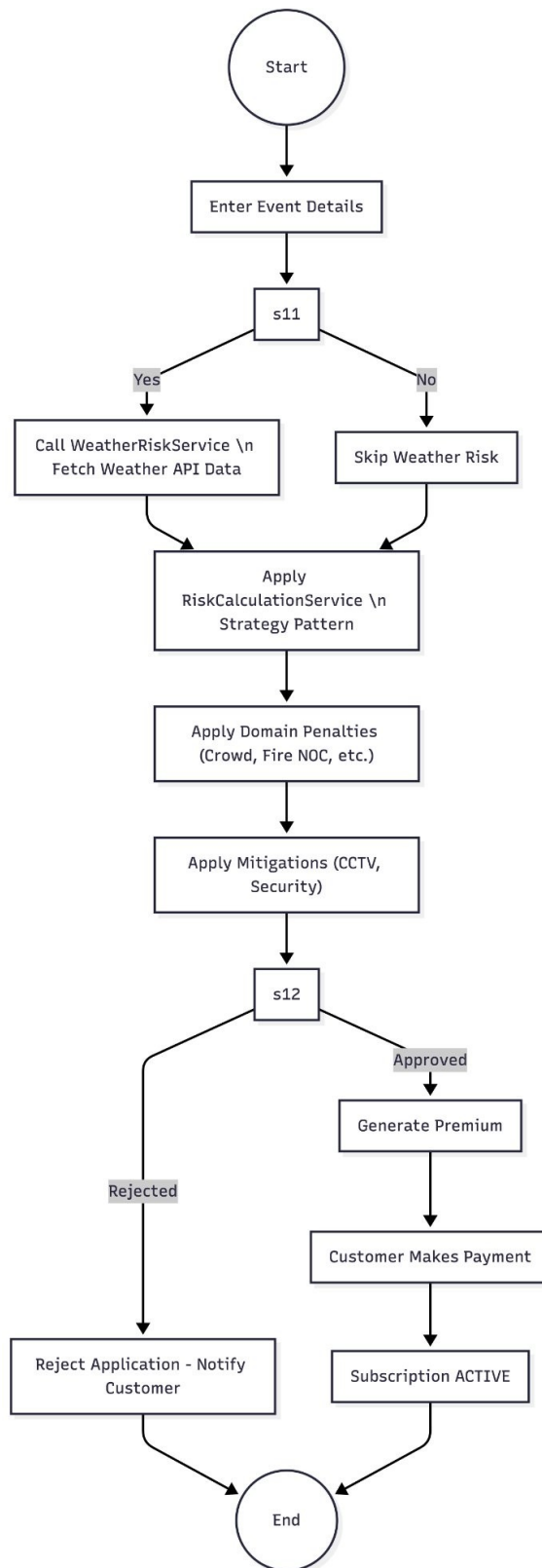


Figure 4 — Activity Diagram — Risk Assessment & Subscription

5. Activity Diagram — Claims Filing & Adjudication

Models the post-incident claims workflow. Decision node s9 validates the claim; s10 branches on the Claims Officer's determination. Approved path triggers payout entry and RESOLVED status; rejected path captures a mandatory rejection reason.

Figure 5 — Activity Diagram — Claims Filing & Adjudication

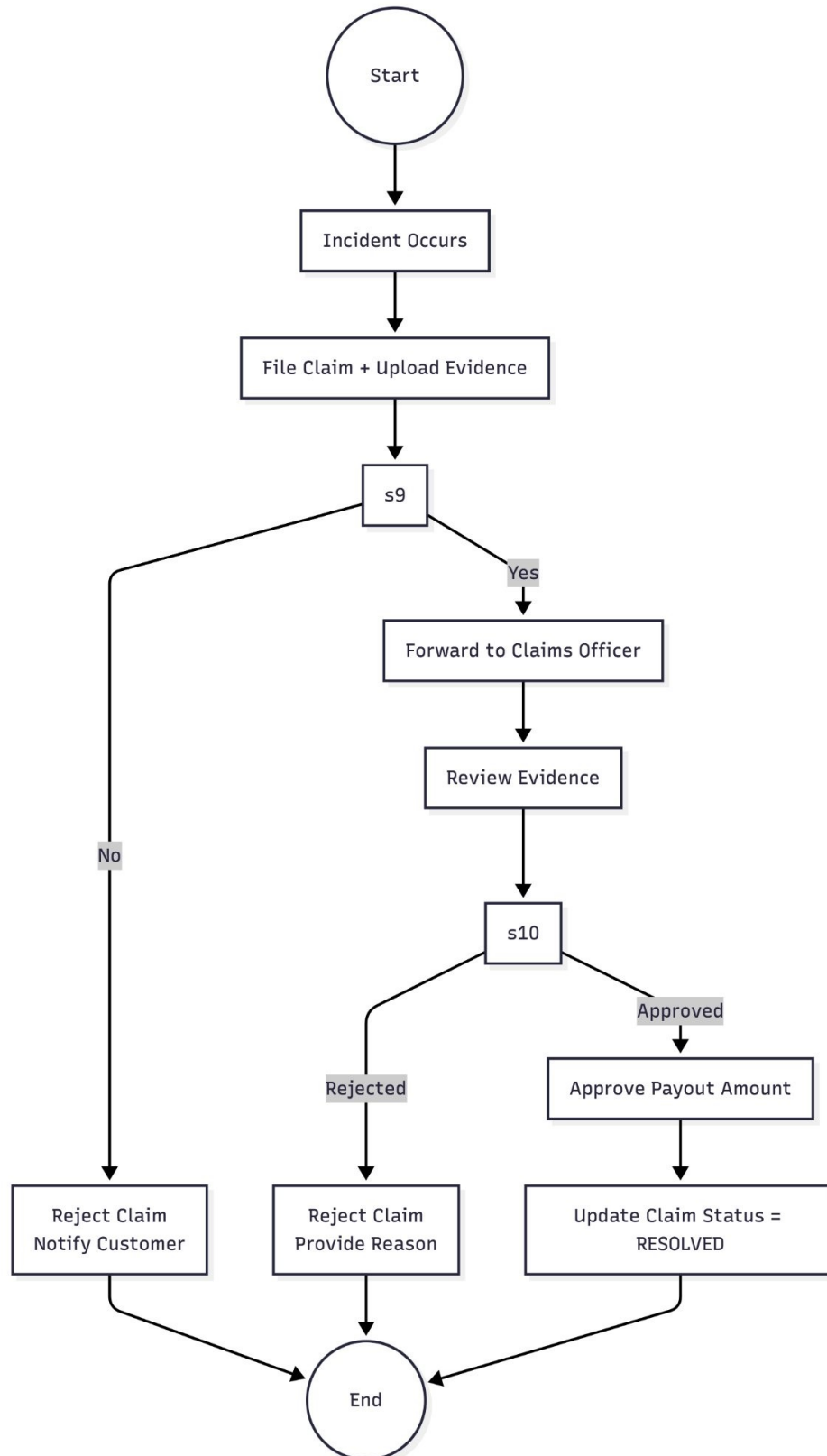


Figure 5 — Activity Diagram — Claims Filing & Adjudication

6. Deployment Diagram

Illustrates Developer Workstation → CI/CD (Jenkins/GitHub Actions with JUnit 5 and Karma/Jasmine parallel test suites) → Production Web Server (Nginx → TLS → Spring Boot API Gateway → H2 persistent file-based storage). WeatherAPI.com shown as third-party integration.

Figure 6 — Deployment Diagram

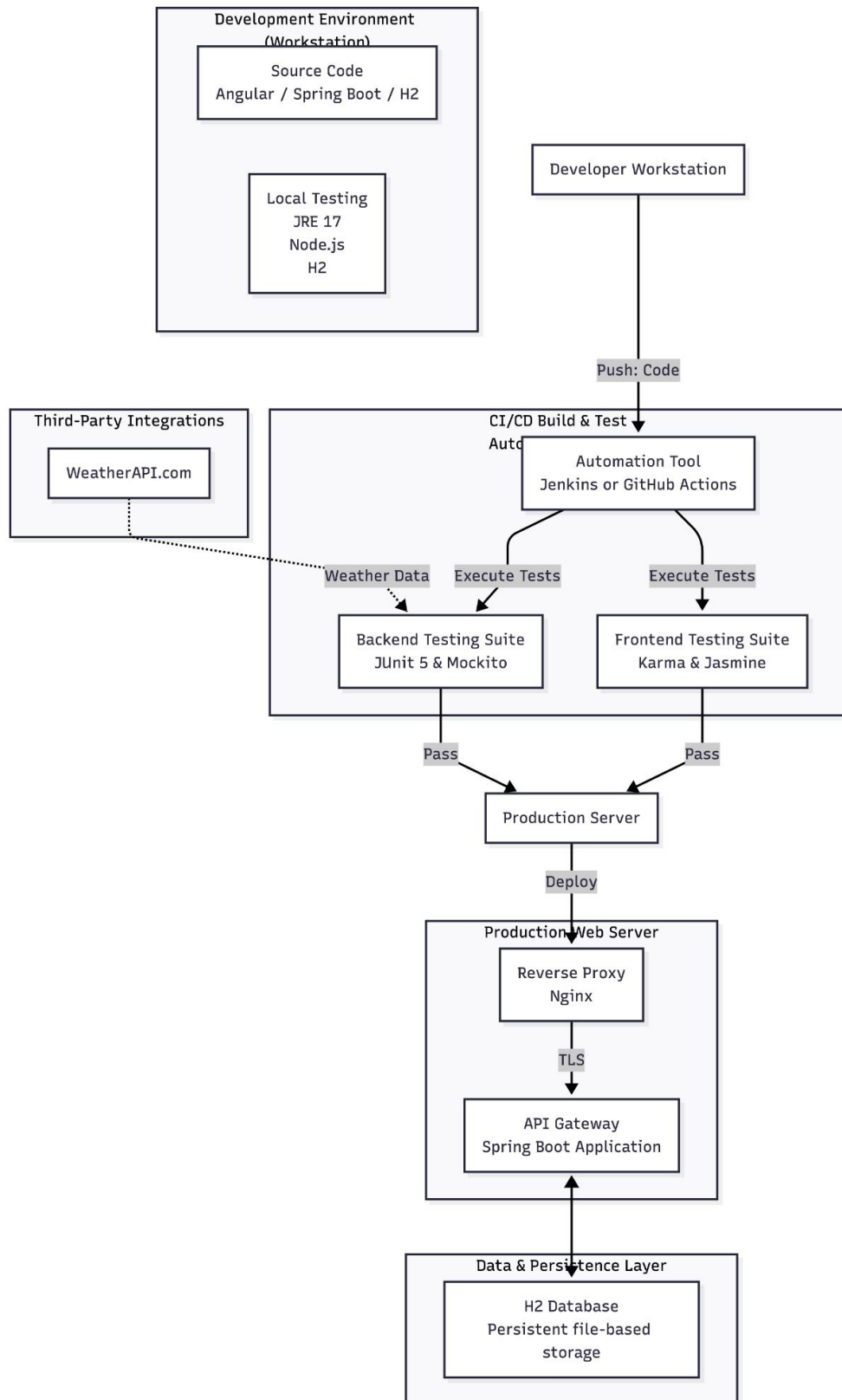


Figure 6 — Deployment Diagram